HOSPITAL PATIENT DATA ANALYSIS

Capstone Project Report
Data Fundamentals and Data Warehousing

Student Details	
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Subject	Data Fundamentals and Data Warehousing

Project Title:	Hospital Patient Data Analysis
Tech Stack:	Python, SQLite3, Excel (openpyxl), Matplotlib
Dataset:	30 Patient Records | 6 Attributes

1. Problem Statement

A hospital maintains records of patients including age, gender, diagnosed disease, assigned doctor, and treatment cost. The goal is to analyze this data to identify disease patterns, doctor workload, and cost distribution to support better hospital management decisions.

- Which diseases are most common?
- Which doctor treats the most patients?
- What is the average treatment cost per disease?
- How are patients distributed across age groups?

2. Dataset

30 patient records (expanded from 5-row sample) with 6 columns:

Column	Type	Description	Example
patient_id	Integer	Unique ID	1, 2...
age	Integer	Patient age	22–67
gender	Text	Male / Female	Male
disease	Text	Diagnosed disease	Diabetes
doctor	Text	Assigned doctor	Dr Shah
treatment_cost	Float	Cost in Rs	1100–25000

Sample Records:

ID	Age	Gender	Disease	Doctor	Cost (Rs)
1	45	Male	Diabetes	Dr Shah	5,000
2	30	Female	Fever	Dr Patil	2,000
3	55	Male	Heart Disease	Dr Shah	20,000
4	22	Female	Flu	Dr Kulkarni	1,500
5	40	Male	Diabetes	Dr Shah	4,500

3. SQL Tasks

Task 1 — Display All Records

```
SELECT * FROM patients;
```

SQL Task 1 - All Patient Records (30 Total)

ID	Age	Gender	Disease	Doctor	Cost (Rs)
1	45	Male	Diabetes	Dr Shah	5,000
2	30	Female	Fever	Dr Patil	2,000
3	55	Male	Heart Disease	Dr Shah	20,000
4	22	Female	Flu	Dr Kulkarni	1,500
5	40	Male	Diabetes	Dr Shah	4,500
...	...	...	...	...	...
28	34	Female	Flu	Dr Patil	1,100
29	61	Male	Diabetes	Dr Kulkarni	5,100
30	27	Female	Fever	Dr Shah	1,750

Output: All 30 records displayed

Task 2 — Count by Disease

```
SELECT disease, COUNT(*) AS patient_count FROM patients GROUP BY disease ORDER BY
patient_count DESC;
```

SQL Task 2 - Patient Count by Disease

Disease	Patient Count
Diabetes	9
Heart Disease	8
Fever	7
Flu	6

Output: Diabetes=9, Heart Disease=8, Fever=7, Flu=6

Task 3 — Average Treatment Cost

```
SELECT disease, ROUND(AVG(treatment_cost),2) AS avg_cost FROM patients GROUP BY disease ORDER
BY avg_cost DESC;
```

SQL Task 3 - Average Treatment Cost

Disease	Avg Cost (Rs)	Min Cost (Rs)	Max Cost (Rs)
Heart Disease	21,500.00	18,000	25,000
Diabetes	5,177.78	4,500	6,000
Fever	1,964.29	1,500	2,200
Flu	1,350.00	1,100	1,600

Output: Heart Disease=Rs.21,500 | Diabetes=Rs.5,178 | Fever=Rs.1,964 | Flu=Rs.1,350

Task 4 — Doctor with Most Patients

```
SELECT doctor, COUNT(*) AS total_patients FROM patients GROUP BY doctor ORDER BY
total_patients DESC;
```

SQL Task 4 - Doctor Treating Most Patients

Doctor	Total Patients	Avg Cost (Rs)	Total Revenue (Rs)
Dr Shah (TOP)	13	11,250	1,46,250
Dr Patil	9	6,044	54,400
Dr Kulkarni	8	4,975	39,800

Output: Dr Shah=13 patients | Dr Patil=9 | Dr Kulkarni=8

4. Python Tasks

Task 1 — Load Dataset

```
import pandas as pd df = pd.read_csv("patients.csv") print(df.head())
```

Result: 30 records loaded with 6 columns confirmed.

Task 2 — Disease Frequency

```
disease_freq = df['disease'].value_counts() print(disease_freq)
```

Result: Diabetes (9) is the most frequent disease.

Task 3 — Treatment Cost Statistics

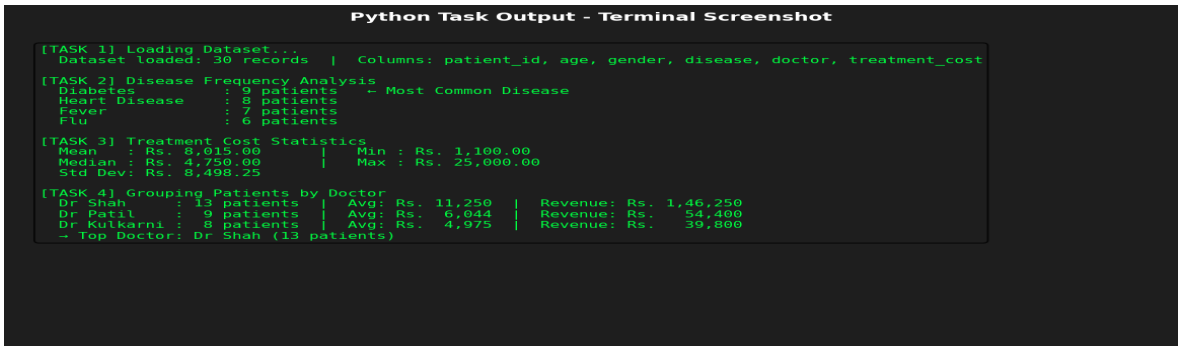
```
print(df['treatment_cost'].describe()) print(df.groupby('disease')['treatment_cost'].mean())
```

Result: Mean=Rs.8,015 | Median=Rs.4,750 | Min=Rs.1,100 | Max=Rs.25,000

Task 4 — Group by Doctor

```
doctor_group = df.groupby('doctor').agg( Total=('patient_id','count'),
Revenue=('treatment_cost','sum') ).sort_values('Total', ascending=False) print(doctor_group)
```

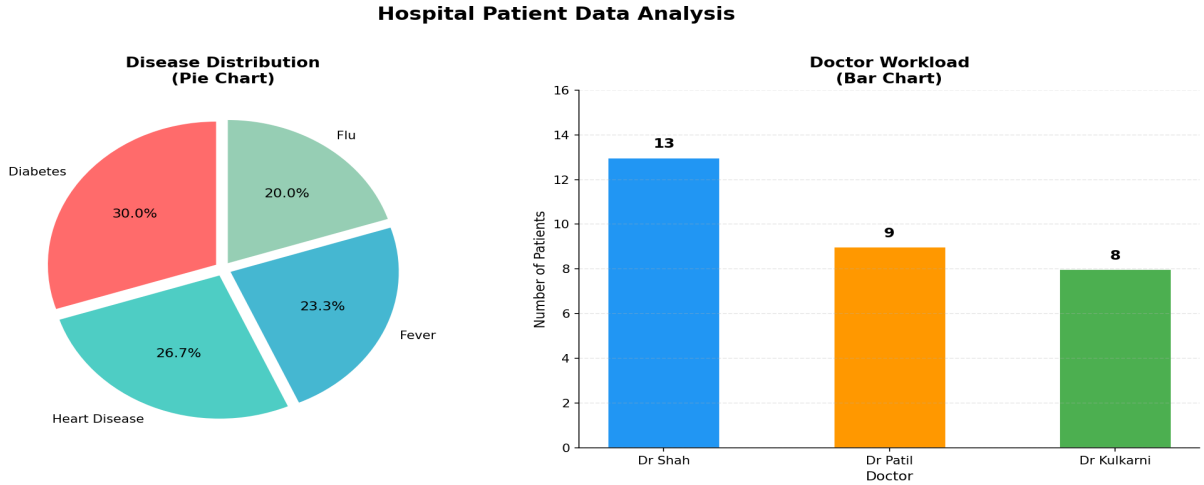
Result: Dr Shah leads with 13 patients and Rs.1,46,250 total revenue.



Screenshot: Python Terminal Output

5. Visualizations

Pie chart (disease distribution) and bar chart (doctor workload) created using Matplotlib.



Charts: Disease Distribution (Pie) | Doctor Workload (Bar)

6. Excel Tasks

Excel workbook created with 3 sheets: Patient Data, Disease Summary (pivot), Doctor Workload (pivot + charts).

Disease	Patient Count	Avg Cost (Rs)
Diabetes	9	5,177.78
Heart Disease	8	21,500.00
Fever	7	1,964.29
Flu	6	1,350.00

Pivot Table: Rows=Disease | Values=Patient Count

7. Key Findings

Most Common Disease	Diabetes — 9 patients (30% of total)
Costliest Disease	Heart Disease — avg Rs. 21,500 per patient
Top Doctor	Dr Shah — 13 patients (43% of all patients)
Lowest Cost Disease	Flu — avg Rs. 1,350 per patient
Age Group	30–45 age group has the most patients (13)
Overall Avg Cost	Rs. 8,015 per patient across all diseases

8. Files Submitted

File	Description
patients.csv	Dataset — 30 patient records
hospital_queries.sql	SQL — table creation + 5 queries
hospital.db	SQLite3 database
hospital_analysis.py	Python — 4 tasks + pie & bar charts
hospital_analysis.xlsx	Excel — 3 sheets with pivot & charts
Hospital_Patient_Data_Analysis_Report.pdf	This project report